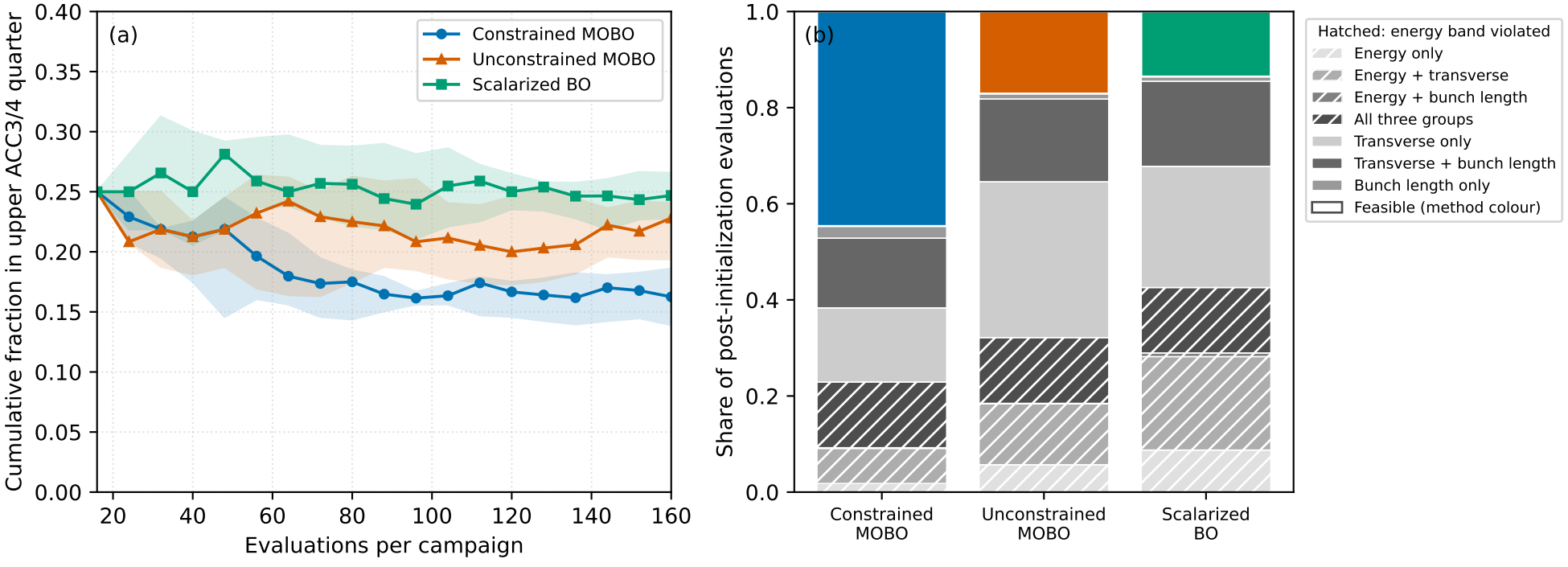
(a)
Constrained MOBO
Unconstrained MOBO
Scalarized BO
Cumulative fraction in upper ACC3/4 quarter
Evaluations per campaign
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
20
40
60
80
100
120
140
160
(b)
Share of post-initialization evaluations
0.0
0.2
0.4
0.6
0.8
1.0
Constrained MOBO
Unconstrained MOBO
Scalarized BO
Hatched: energy band violated
Energy only
Energy + transverse
Energy + bunch length
All three groups
Transverse only
Transverse + bunch length
Bunch length only
Feasible (method colour)